

1. a) $\int_4^6 \frac{x-3}{x^2-3x+2}\,dx = \int_4^6 \frac{-1}{(x-2)} + \frac{2}{(x-1)}\,dx$

$A(x-1) + B(x-2) = x-3$

let $x=1$: $-B = -2$

$B = 2$

let $x=2$: $A = -1$

$$\left[-\ln|x-2| + 2\ln|x-1|\right]_4^6 = \left[\ln(x-1)^2 - \ln(x-2)\right]_4^6$$

$$\left[\ln\left(\frac{(x-1)^2}{x-2}\right)\right]_4^6 = \ln\left(\frac{5^2}{4}\right) - \ln\left(\frac{3^2}{2}\right) = \ln\left(\frac{25/4}{9/2}\right)$$

$$= \ln\frac{25}{18}$$

b) $\frac{1}{2}\ln\left(\frac{25}{18}\right) = \ln\frac{25^{1/2}}{18^{1/2}} = \ln\frac{5}{3\sqrt{2}} = \ln\frac{5\sqrt{2}}{6}$

2. a) $\frac{d}{dx}(\arccos x) = \frac{-1}{\sqrt{1-x^2}}$

$y = \arccos x$
$x = \cos y$ $\quad x^2 = \cos^2 y$ $\quad \sin^2 y = 1 - \cos^2 y = 1 - x^2$

$\frac{dx}{dy} = -\sin y$ $\quad \frac{dy}{dx} = \frac{-1}{\sin y}$ $\quad \sin y = \sqrt{1-x^2}$

$\frac{dy}{dx} = \frac{-1}{\sqrt{1-x^2}}$

b) $m = \frac{-1}{\sqrt{1-\left(\frac{\sqrt{3}}{2}\right)^2}} = -2$ $\quad \left(\frac{\sqrt{3}}{2}, \frac{\pi}{6}\right)$

$y - \frac{\pi}{6} = -2\left(x - \frac{\sqrt{3}}{2}\right)$

$y - \frac{\pi}{6} = -2x + \sqrt{3}$

$y + 2x - \frac{\pi}{6} - \sqrt{3} = 0$

3. a) the function is undefined at the upper limit.

b) $\int_0^1 \frac{1}{\sqrt{1-x}}\,dx = \int (1-x)^{-1/2}\,dx = 2(1-x)^{1/2}$